# F1 Car Race Prediction

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## Objective

Our goal is to develop a **predictive model** that would evaluate key factors on which the **performance** of an F1 driver depends and give the **probability of winning** future races by **calculating the points table** for said race with given track difficulty and other factors. We have **trained** on data from **2011- 2017** and then trained and tested a **Multivariable Linear Regression model** by simulating it to understand the correlation of factors affecting the race-winning probability, and then **tested** our results with the data of **2018-2020** (both inclusive).

## Theory

1) **Multivariable Linear Regression** is a statistical method used to model the relationship between **one dependent** variable and **two or more independent** variables. For instance, suppose we wish to forecast a student's final exam grade based on the number of hours they studied, the amount of sleep they received the night before, and the number of classes they attended throughout the semester. These three variables—study hours, sleep hours, and class attendance—are referred to as independent variables, whereas the exam score is the dependent variable. The model gives a weight (coefficient) to each variable, illustrating how much of an effect it has on the final mark. Assume after running data from previous students, we obtain a formula such as: score = 20 + 5 × (study hours) + 2 × (sleep hours) + 1 × (classes attended). This indicates that every additional hour of study contributes 5 points to the estimated score, and so forth. If a student has studied for 6 hours, slept for 7 hours, and attended 25 classes, the model would estimate that they would have scored 89. This is how multivariable linear regression assists us in making meaningful predictions by consolidating the influence of multiple variables into one equation.

The generalised equation for the same is: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots\ldots \beta_n x_n + \epsilon$

$$E[Y_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik}$$

If we let $B_0, B_1, \ldots, B_k$ denote estimators of $\beta_0, \ldots, \beta_k$, then the sum of the squared differences between the $Y_i$ and their estimated expected values is

$$\sum_{i=1}^{n} (Y_i - B_0 - B_1 x_{i1} - B_2 x_{i2} - \cdots - B_k x_{ik})^2$$

The least squares estimators are those values of $B_0, B_1, \ldots, B_k$ that minimize the foregoing.

Before solving the normal equations, it is convenient to introduce matrix notation. If we let

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \qquad \mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

then $\mathbf{Y}$ is an $n \times 1$, $\mathbf{X}$ an $n \times p$, $\boldsymbol{\beta}$ a $p \times 1$, and $\mathbf{e}$ an $n \times 1$ matrix where $p \equiv k + 1$.

The multiple regression model can now be written as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

In addition, if we let

$$\mathbf{B} = \begin{bmatrix} B_0 \\ B_1 \\ \vdots \\ B_k \end{bmatrix}$$

be the matrix of least squares estimators, then the normal Equations 9.10.1 can be written as

$$\mathbf{X}'\mathbf{X}\mathbf{B} = \mathbf{X}'\mathbf{Y} \qquad (9.10.2)$$

where $\mathbf{X}'$ is the transpose of $\mathbf{X}$.

Thus,

$$\mathbf{B} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \qquad (9.10.3)$$

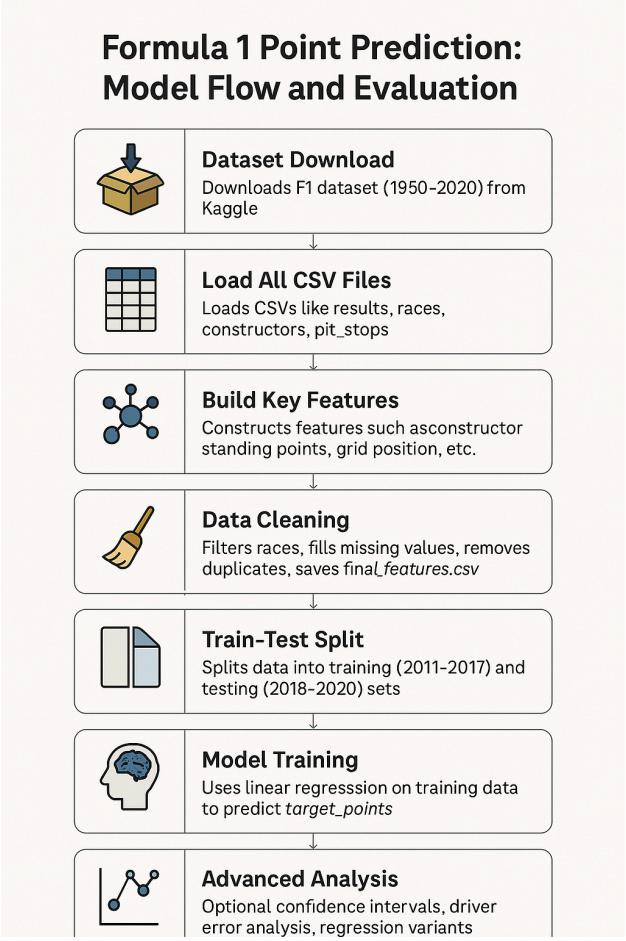
2) The R-squared value (denoted as $R^2$) is a statistical measure that explains how well the independent variables in a regression model account for the variability in the dependent variable. Basically, it tells us how well the model's predictions match the actual data. The value of $R^2$ ranges between 0 and 1. An $R^2$ of 1 means the model perfectly predicts the outcome, while an $R^2$ of 0 means the model does not explain any of the variation in the target variable

## Factors

| Factor | Description |
|---|---|
| Pit Stop Time | The average time a team takes to complete a pit stop in a race. This can impact the overall race time and the total points scored. |
| Grid Position (Current Skill Level) | The starting grid position is determined by the qualifying race, which happens just before the actual race on the same track. A better grid position often leads to a higher finishing position and more points. |
| Constructor Standing | The current ranking of a constructor (team) in the overall championship. A higher position could indicate better performance and more points. |
| Absolute Skill (Cumulative Points) | The driver's skill, measured by average finishing position across multiple races or points accumulated, with higher weight given to better performances. |
| Weather | The weather conditions during the race (e.g., rain, dry, temperature). Weather can heavily influence performance and race strategy. |
| Track Difficulty | |
| Avg Pit Stop Rate (Avg Number of Pit Stops per Race) | The average number of pit stops made by the driver/team in a race. This may correlate with track difficulty and strategy efficiency. |
| DNF Rate (Did Not Finish Rate) | The rate at which a driver or team fails to finish the race. A higher DNF rate can indicate greater |

| | |
|---|---|
| | difficulty or poor reliability, reducing total points scored. |

## Working




Dataset - [Formula 1 World Championship (1950 - 2024)]

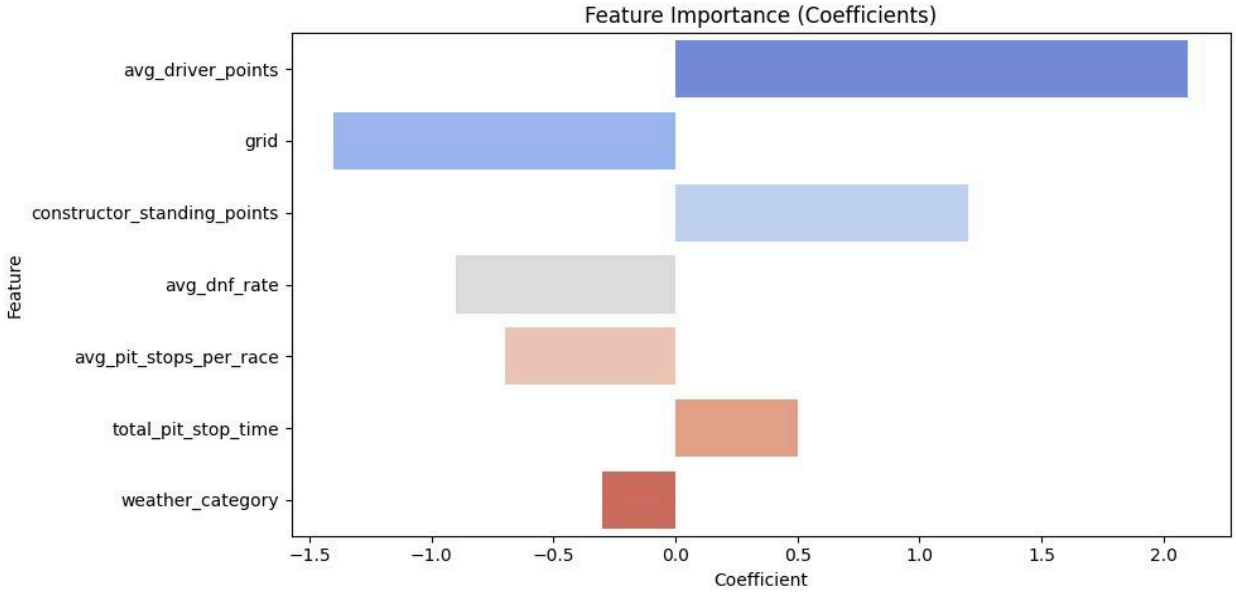
Simulation: IEOR.ipynb

## Results

1) The coefficients of the equation, intercept, mean squared error and mean average error are:

```
Coefficients: [-9.20592238e-03  4.96580090e-04 -6.01452365e-05 -1.79455428e-02
  1.97859773e-02 -1.46796520e-01  8.93535965e-01  5.92477194e-01
  1.73625838e-01 -2.58101876e-01]
Intercept: 9.274458825938513
Mean Squared Error: 26.712283424040038
MAE: 3.6163975335490393
```

2) Feature Importance Bar Chart: It displays how important each feature is in predicting the target variable. It displays how important each feature is in predicting the target variable.If a feature has a high importance score, it suggests that this feature significantly influences the model's prediction. Interesting thing to note is the pit stop time contributes oppositely as compared to reality.



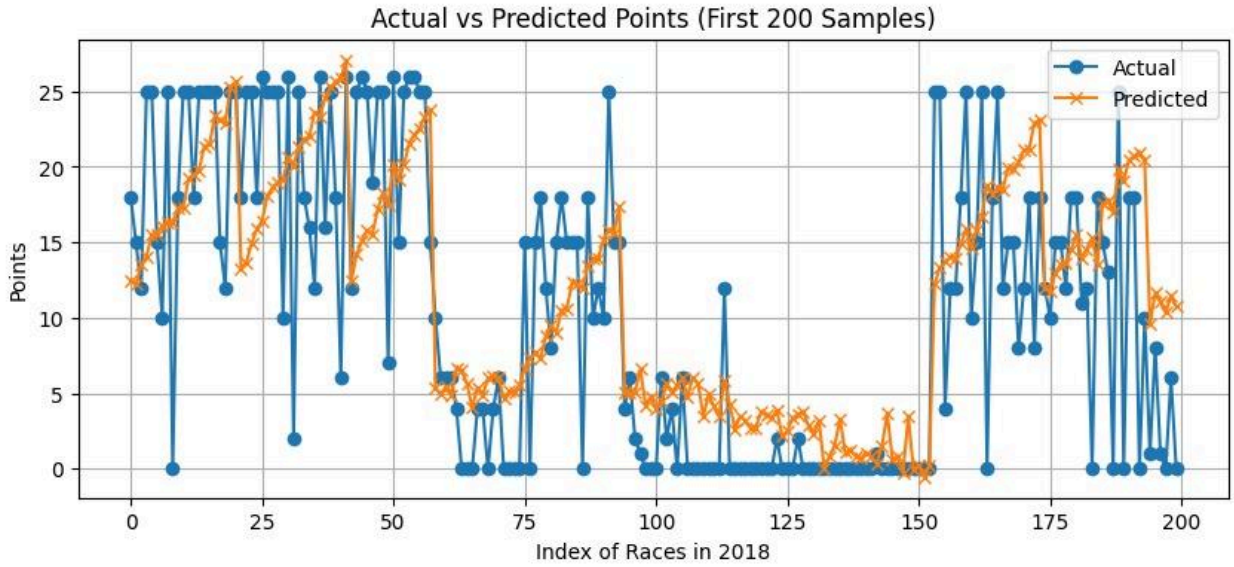

3) The R-squared value by our model for the given dataset is **0.5**. However, when we use the model to test data after 2020, the R-squared values decrease.

| | train_years | test_years | r2 | MAE | RMSE |
|---|---|---|---|---|---|
| 9 | 2011–2017 | 2018–2020 | 0.503935 | 3.616398 | 5.168393 |
| 6 | 2011–2016 | 2017–2020 | 0.501746 | 3.645085 | 5.184405 |
| 3 | 2011–2015 | 2016–2020 | 0.484995 | 3.693432 | 5.238132 |
| 7 | 2011–2016 | 2017–2021 | 0.474549 | 3.806251 | 5.325433 |
| 10 | 2011–2017 | 2018–2021 | 0.474116 | 3.774878 | 5.324390 |
| 12 | 2011–2018 | 2019–2020 | 0.461191 | 3.812677 | 5.387778 |
| 4 | 2011–2015 | 2016–2021 | 0.458195 | 3.859160 | 5.379633 |
| 8 | 2011–2016 | 2017–2022 | 0.452448 | 3.909519 | 5.457636 |
| 11 | 2011–2017 | 2018–2022 | 0.450173 | 3.906425 | 5.470647 |
| 0 | 2011–2014 | 2015–2020 | 0.439402 | 3.900838 | 5.468516 |
| 13 | 2011–2018 | 2019–2021 | 0.436950 | 3.959136 | 5.511197 |
| 5 | 2011–2015 | 2016–2022 | 0.435584 | 3.959479 | 5.513713 |
| 14 | 2011–2018 | 2019–2022 | 0.414621 | 4.086627 | 5.653366 |
| 1 | 2011–2014 | 2015–2021 | 0.410397 | 4.072219 | 5.613946 |
| 18 | 2011–2020 | 2021–2021 | 0.406584 | 4.146852 | 5.664485 |
| 16 | 2011–2019 | 2020–2021 | 0.391473 | 4.265367 | 5.748974 |
| 2 | 2011–2014 | 2015–2022 | 0.385232 | 4.171921 | 5.753358 |
| 19 | 2011–2020 | 2021–2022 | 0.382322 | 4.292278 | 5.841188 |
| 15 | 2011–2019 | 2020–2020 | 0.381162 | 4.324474 | 5.813124 |
| 17 | 2011–2019 | 2020–2022 | 0.377258 | 4.335032 | 5.855518 |

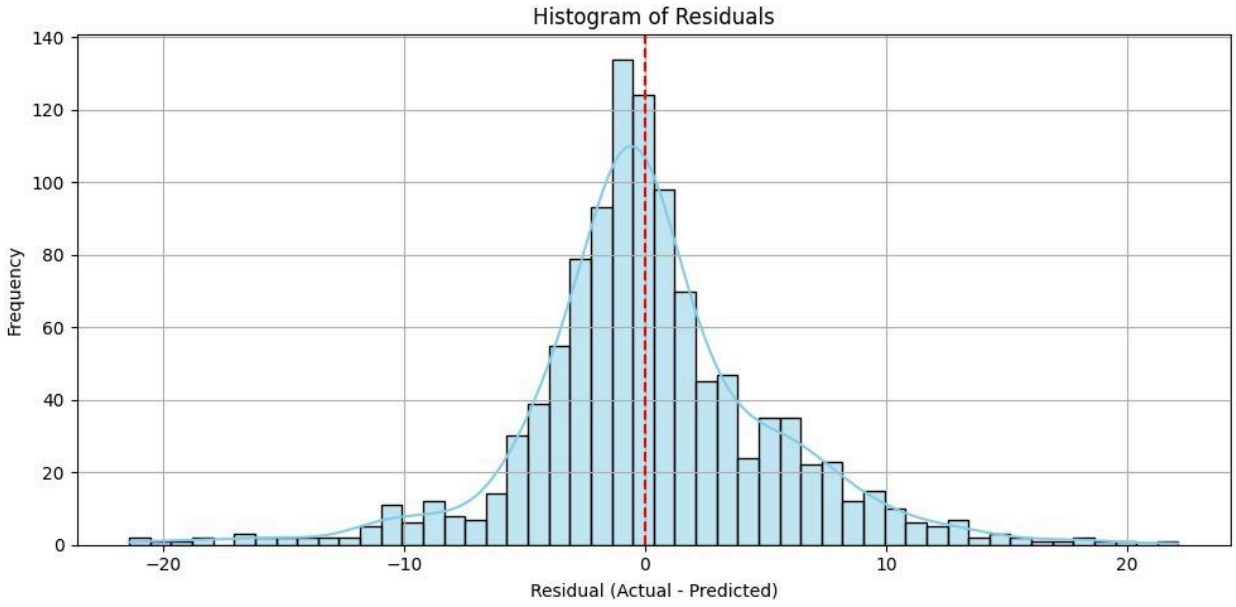
Cause of decrease in R squared: Before the execution of the cost cap in F1 racing, top teams such as Mercedes, Ferrari, and Red Bull used to spend between $220 million to $250 million annually on car performance-related expenditures. In contrast, smaller teams had lower budgets, leading to notable disparities in competitiveness.

In response to financial sustainability concerns and to promote a more level playing field, Formula 1 introduced a cost cap starting in the 2021 season. The cap was set at $145 million for 2021, with plans to reduce it to $140 million in 2022 and further to $135 million from 2023 onwards. This cap specifically targets expenditures related to car performance and excludes certain costs such as driver salaries, marketing expenses, and the salaries of the top three personnel. This caused our R-squared to get reduced from 0.5 to 0.36 when we predicted the race result after 2021.

4) Here is a graph of our first 200 predictions and their actual values when the model is tested from 2018 onwards.

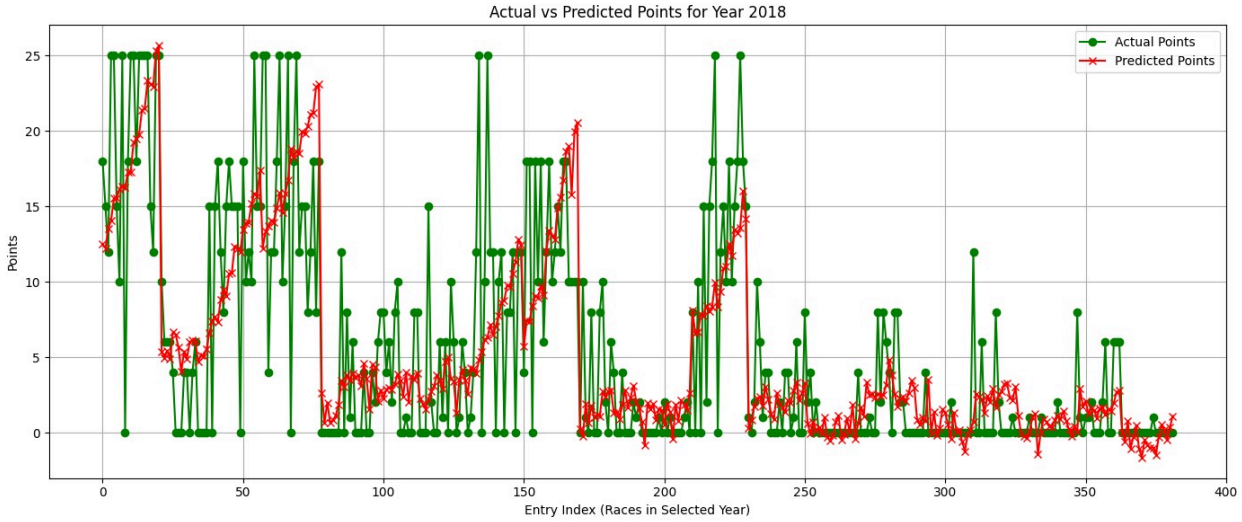


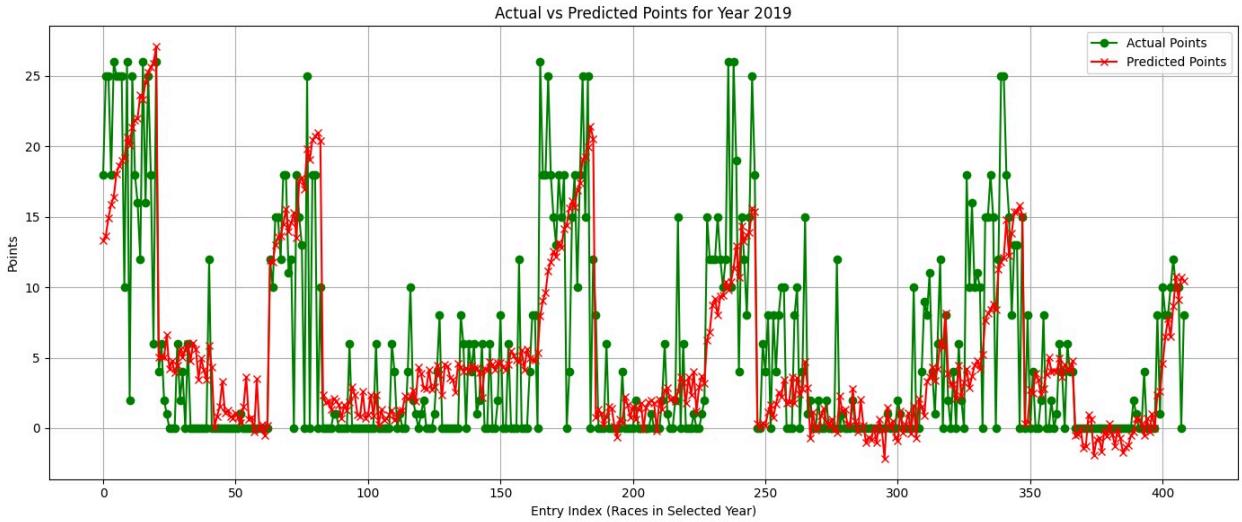

5) Checking our errors: Histogram of the Residuals shows the distribution of the residuals (errors). It should ideally resemble a normal distribution. A normal-shaped distribution of residuals indicates that the model's errors are random and the assumptions of linear regression hold. Here is a histogram of our errors, which is almost normally distributed.

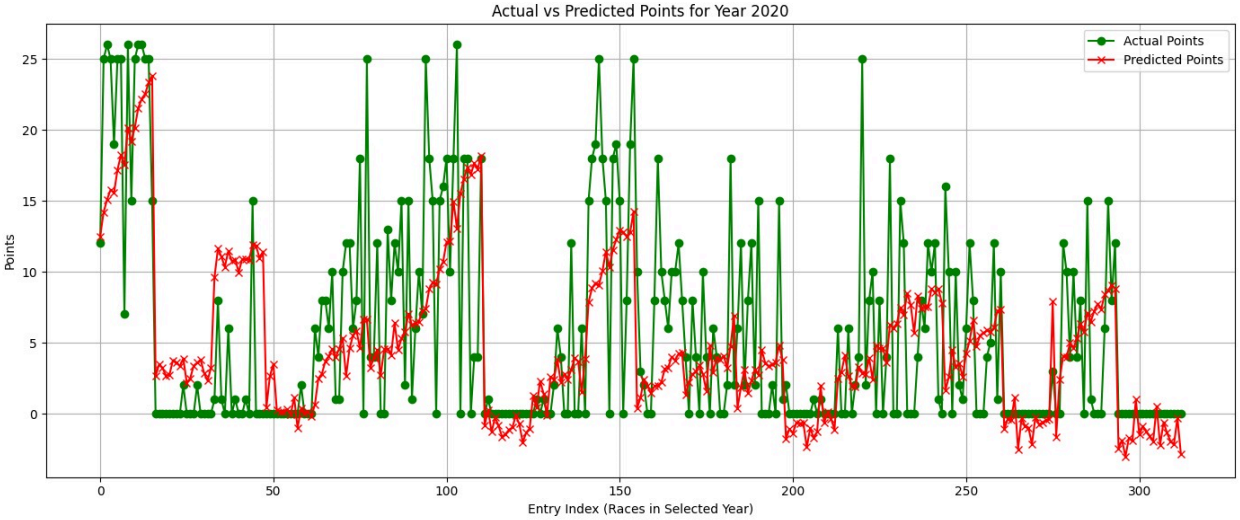


Linear regression is based on a set of assumptions to give unbiased and consistent estimates for the model coefficients. One of these assumptions is that the errors (residuals) must be normally distributed.

# Resulting Graphs




Actual vs Predicted Points for Year 2020
Actual Points
Predicted Points
Points
Entry Index (Races in Selected Year)
25
20
15
10
5
0
0
50
100
150
200
250
300